

Stochastic Simulation

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1 Introduction

In this report, we simulate the use of CT scanners in the radiology department. The department has three types of patients: emergency patients, inpatients and outpatients. We analyze it as a stochastic system, since patient arrivals and scan durations are random. Discrete-event simulation is used to model the problem, because relevant changes in the system occur at discrete event times, such as patient requests, patient arrivals, scan starts and scan completions.

First, the report describes the system with CT scanners and analyzes the performance indicators. After that, we analytically examine the capacity of the scanners. Subsequently, the discrete-event simulation and its statistical analysis are demonstrated. Finally, the simulation results are presented. This assignment aims to evaluate the current system's capacity and inspect performance indicators such as CT scanner utilization, waiting time, and access time.

2 Problem Description in Words

The radiology department serves three types of patient, each with a different arrival rate and service rules: emergency patients, inpatients and outpatients. There are two CT scanners available on weekdays 8:00 to 16:00, and one that works the rest of the hours. This report only considers fixed current policy. The scan duration is uniformly distributed between 10 and 19.

Emergency patients directly arrive at the departments, and are served as soon as possible and their arrival is modelled by Poisson process with rate ($\lambda_E = 24$) *per day*.

Inpatients request a scan before arrival. Requests follow Poisson distribution with rates 21 and 6 requests during office hours and outside office hours respectively. They wait until they get a call and after that they are transported to CT with transport time being uniformly distributed between 9 and 15 minutes.

Outpatients request an appointment during office hours according to a Poisson process with rate ($\lambda_O = 23$) per weekday. The earliest an outpatient can be scheduled is the next day. Four slots per hour are offered between 8:00 and 12:00, and three slots per hour between 12:00 and 16:00. If there are no slots available in the same week, the outpatient will be placed on a waiting list and cared for on Friday night of the next week. The probability of an outpatient to show up is $p_s = 0.84$.

Emergency patients are always served first. If no emergency patient is present, inpatients and outpatients are served in order of arrival. The waiting room has three chairs. If more than three patients are waiting, the remaining people are considered to be waiting outside the waiting room.

This assignment's performance indicators are CT scanner utilization during and outside office hours, outpatient access time, waiting times for emergency patients and outpatients, the fraction of CT patients waiting outside the waiting room, and the percentage of office-hour inpatient requests not scanned before 16:00 on the same weekday.

3 Analytical Capacity Check

We want to find out if two CT scanners are sufficient for treating all patients. This is checked analytically: the expected CT scanner workload is compared to scanner capacity.

For all patient types, the CT scan duration follows $U(10, 19)$. Let S denote the scan duration. The expected scan duration is

$$E[S] = \frac{10 + 19}{2} = 14.5 \text{ minutes} \approx 0.2417 \text{ hours}$$

Let N_E , N_I and N_O denote the number of expected weekly scans of emergency patents, inpatient and outpatient. Let W_E , W_I and W_O be their expected weekly workloads for a week.

Emergency patients arrival is a Poisson process with rate $\lambda_E = 24$ per day and $N_E = 24 \cdot 7 = 168$.

The expected emergency workload is

$$W_E = N_E E[S] = 168 \cdot 0.2417 \approx 40.60$$

hours of workload. Inpatients have on average 21 requests during office hours and 6 requests arrive outside office hours. We assume that there are 0 requests during weekends. Therefore,

$$N_I = 5(21 + 6) = 135,$$

and

$$W_I = N_I E[S] = 135 \cdot 0.2417 \approx 32.63$$

Outpatients requests have a rate of $\lambda_O = 23$ per weekday with propabability to show up $p_s = 0.84$. Therefore,

$$N_O = 5 \cdot 23 \cdot 0.84 = 96.6,$$

and the expected hours of workload:

$$96.6 \cdot 0.2417 \approx 23.35$$

The total expected scanner workload W is

$$W = W_E + W_I + W_O = 40.60 + 32.63 + 23.35 = 96.58$$

Let C_{office} and C_{outside} denote scanner capacity during office and outside office hours respectively. Two scanners are available from 8:00 to 16:00 on weekdays, so

$$C_{\text{office}} = 5 \cdot 8 \cdot 2 = 80$$

hours per week. Outside office hours, one scanner is available. There are 168 hours in a week and 40 of them are office hourse. Therefore,

$$C_{\text{outside}} = 168 - 40 = 128$$

hours per week. Hence the total capacity C :

$$C = C_{\text{office}} + C_{\text{outside}} = 80 + 128 = 208$$

Since $W < C$, we can conclude that the department's workload is smaller than its capacity.

4 Simulation Model

The CT department is modelled as a discrete-event simulation.

We make the following assumptions. Emergency patients have priority over all the other patients, while inpatients and outpatients are served in the order of arrival. A scan that has started before 16:00 is allowed to finish after 16:00. Scan duration follows $U(10, 19)$, i.e., the scan duration is distributed uniformly between 10 and 19 minutes. We ignore PAS patients so they do not occupy the chairs inside the waiting room.

The simulation state consists of the emergency queue for the emergency patients, the normal queue occupied by both the inpatients and the arrived outpatients, the outpatient waiting list, the scanner states, the outpatient appointment schedule, and the statistics counters.

The model keeps track of the number of patients who arrive when all the three waiting-room chairs are occupied, the number of office-hour inpatient requests, the number of office-hour inpatients who are not scanned before 16:00 on the same weekday, outpatient access times, emergency waiting times, and scanner utilisation during and outside office hours.

The event types used in the model are emergency arrival, outpatient request, outpatient appointment arrival, inpatient arrival, scan completion, Friday waiting-list scheduling, and capacity change.

At an emergency arrival event, a patient goes directly to the CT department. A patient is added to the emergency queue, and the waiting room counter is increased by one. If the three chairs in the waiting room are already taken, the patient is counted as waiting outside.

At an inpatient request event, the inpatient is added to the normal queue. If the request happens during the office hours, the model records it as an office-hour inpatient request and stores the same-day deadline of 16:00. The model records a waiting-room arrival and checks whether the patient has to wait outside. Then the model checks whether a scan can start. Finally, the next inpatient arrival is scheduled.

At an outpatient request event, an outpatient requests an appointment. The model searches for the earliest available appointment slot in the same week, but not on the same day as the request. In the morning, from 08:00 to 12:00, four outpatient slots are available per hour. In the afternoon, from 12:00 to 16:00, three outpatient slots are available per hour. If a slot is found, the appointment is scheduled, and the outpatient access time is recorded as the number of days between the request and the appointment. If no slot is available in the same week, the outpatient is added to the outpatient waiting list. The next outpatient request is then scheduled.

At a scan completion event, the scan of a patient ends, and the scanner becomes free. The number of completed patients is increased. The model then checks whether another patient can start scanning immediately. If an emergency patient is waiting, that patient is selected first. Otherwise, the first patient in the normal queue is selected. When a scan starts, the patient's waiting time is recorded. For emergency and outpatient patients, this is the time between arrival in the waiting room and the start of the scan. For inpatients with office-hour requests, the model checks whether the scan starts before 16:00 on the same weekday.

At a Friday waiting-list scheduling event, all outpatients on the waiting list are considered in the order in which they entered the waiting list. The model schedules them into available appointment slots in the next week. If a patient receives an appointment, the access time is recorded, and an outpatient appointment arrival event is scheduled. If no appointment slot is available, the patient remains on the waiting list.

Capacity change event happens when scanner availability changes (at 8:00 and 16:00 on weekdays). The utilisation statistics is renewed, and the model checks if a new scan can start.

Scan completion means that a scan is finished and the scanner is free.

5 Flowchart

The overall structure of the discrete-event simulation model is given in Figure 1.

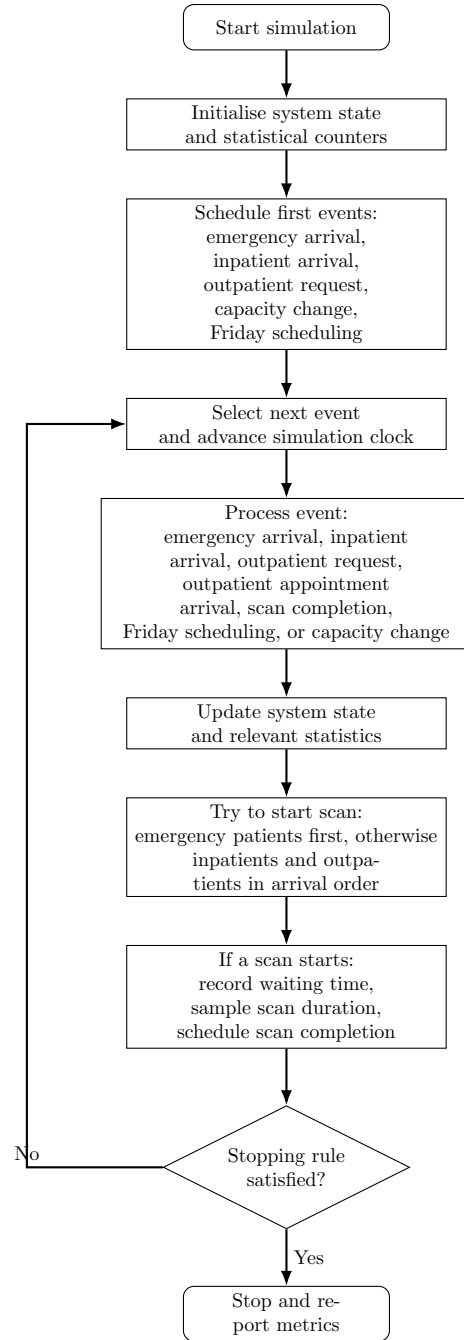


Figure 1: Flowchart of the CT scan department simulation. Events are processed in chronological order. After each event, the system state and statistics are updated, and the model checks whether a new scan can start.

6 Results

6.1 Simulation Results

The simulation results are shown in Table 1.

Table 1: Simulation results (70 replications)

Metric	Mean	95% CI	Half-width	Rel. prec.
Office util.	0.7124	[0.7095, 0.7152]	2.90×10^{-3}	4.07×10^{-3}
Outside util.	0.3368	[0.3350, 0.3385]	1.78×10^{-3}	5.29×10^{-3}
Emerg. wait (min)	3.155	[3.110, 3.200]	4.48×10^{-2}	1.42×10^{-2}
Outpat. wait (min)	2.886	[2.823, 2.950]	6.33×10^{-2}	2.19×10^{-2}
Outpat. access (days)	1.3872	[1.3809, 1.3935]	6.30×10^{-3}	4.54×10^{-3}
Frac. waited outside	0.002796	[0.002532, 0.003061]	2.64×10^{-4}	9.44×10^{-2}
Frac. missed same-day	0.02935	[0.02687, 0.03184]	2.49×10^{-3}	8.47×10^{-2}

The simulation results suggest that the CT department has sufficient capacity. The simulated utilisation is about 0.712 during office hours and 0.337 outside office hours. Both values are below full capacity, although the office-hour period is more heavily used.

The waiting times are relatively low. Emergency patients wait on average about 3.16 minutes, while outpatients wait about 2.89 minutes after arriving at the CT department. The average outpatient access time is about 1.39 days, so outpatients usually receive an appointment slightly more than one day after their request.

Only about 0.280% of CT patients have to wait outside the waiting room, so the three-chair waiting room is usually sufficient. The fraction of office-hour inpatients missing the same-day target is about 2.94%, meaning that most office-hour inpatient requests are scanned before 16:00 on the same weekday. All reported performance measures satisfy the required 10% relative precision.

7 Conclusion

The current CT department appears to have sufficient capacity under the simulated parameter setting. Waiting times are short, outpatient access times are low, and only a small fraction of patients have to wait outside the waiting room. The main pressure point is the office-hour period, where utilization is higher and about 2.94% of office-hour inpatient requests miss the same-day target. Overall, the simulated system performs reasonably well.